

IE 6600

Project 2 - Final Report

Advanced Data Analysis and Visualization with Seaborn

Dataset: NHTSA Vehicle Recalls, 2010-2026

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Data source: data.gov / National Highway Traffic Safety Administration (NHTSA) recall records.

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Abstract

This report uses the NHTSA vehicle recalls dataset to study recall activity in the United States from 2010 through 2026. The original CSV contains 240,137 recall records. Because the same recall campaign can appear multiple times for different makes, models, or model years, the analysis also creates a campaign-level table with 14,912 unique campaigns. The work is completed in Python using pandas for data cleaning, Seaborn for colorful static visualizations, matplotlib for figure export, and SciPy for basic statistical checks.

Three main findings came out of the analysis. First, recall activity is uneven across years: the highest yearly count in this file is 1,094 unique campaigns in 2021, while the lowest yearly count is 305 in 2026. Second, campaign counts are concentrated by make and component; FORD has the highest campaign count, and EQUIPMENT is the most frequent component group. Third, the number of potentially affected units is extremely right-skewed: the median campaign affects 474 units, the 90th percentile reaches 52,374, and the 99th percentile reaches 894,651.

Therefore, log-scale charts are more readable than raw-scale charts.

High-urgency safety instructions are less common than ordinary recall notices but still important. In the campaign-level file, 201 campaigns include Do Not Drive instructions and 94 campaigns include Park Outside instructions. These special flags identify cases where the recall message is not just about future repair scheduling, but about immediate consumer behavior.

Keywords: Seaborn, static visualization, NHTSA, vehicle recalls, campaign-level analysis, transportation safety, exploratory data analysis.

1. Introduction

1.1 Background

Vehicle recalls are an important public-safety process. When a manufacturer or regulator finds that a vehicle, part, or system may create a safety risk or fail to meet a standard, a recall campaign is created. The recall process connects technical defects with consumer-facing actions such as repair notices, replacement parts, inspections, software updates, or urgent warnings. Because recalls can involve thousands or even millions of vehicles, the data is useful for both transportation safety and operations analysis.

The NHTSA recall dataset is especially suitable for a Seaborn visualization project because it includes both categorical and numeric fields. Categorical fields describe the make, model, manufacturer, component, and influence source. Numeric and date fields describe the model year, recall date, and potentially affected population. This mix allows us to build ranked bar charts, time-series charts, distribution plots, heatmaps, and regression plots.

This dataset is also useful because it connects safety issues with operational impact. A recall is not only a technical problem; it also creates work for manufacturers, suppliers, dealers, and customers. Large recalls may require parts planning, repair scheduling, customer communication, warranty processing, and follow-up tracking. For this reason, studying recall data can help identify where recall activity is concentrated and which areas may need stronger monitoring or response planning.

1.2 What we wanted to find out

- Q1. How has unique recall campaign activity changed over time?
- Q2. Which makes and vehicle components appear most often in recall campaigns?
- Q3. How large are recall campaigns, and is the affected-unit distribution balanced or skewed?
- Q4. Which recalls include high-urgency instructions such as Do Not Drive or Park Outside?
- Q5. Does vehicle age at recall notice appear related to recall campaign scale?

1.3 Scope of this report

The report is descriptive and visual. It does not attempt to predict future recalls or judge the quality of any manufacturer. Instead, it focuses on organizing the raw file, avoiding over-counting repeated campaign records, creating clear Seaborn charts, and explaining what each chart suggests.

The scope includes data cleaning, campaign-level summarization, exploratory data analysis, and static visualization using Seaborn. The analysis looks at recall activity over time, top makes, top components, affected-unit distribution, special safety flags, influencing sources, and vehicle age at recall notice. These areas were selected because they connect both safety-related and operations-related questions.

The report also includes basic statistical checks such as skewness, kurtosis, normality testing, and simple correlation analysis. However, it does not include advanced prediction, causal modeling, or manufacturer-level safety ranking. More formal modeling could be added later, but the current report is intentionally focused on interpretation, visualization quality, and reproducibility.

2. Dataset

2.1 Where the data came from

The dataset was downloaded from data.gov and is based on National Highway Traffic Safety Administration recall records. In this file, the recall date range runs from 2010 through 2026. The raw file is large because a recall campaign may be listed across multiple model years or vehicle models. For that reason, using every row as a separate recall would overstate activity. The cleaned analysis therefore uses both the full record-level table and a campaign-level summary table.

2.2 What each row contains

Column	Type	What it stores
CAMPNO	string	Recall campaign number used to connect repeated rows
MAKETXT / MODELTEXT	string	Vehicle make and model involved in the record
YEARTXT	numeric	Vehicle model year
COMPNAME	string	Vehicle component or system involved
MFGNAME	string	Manufacturer name
RCDATE / ODATE	date	Recall report date and owner notification date
Column	Type	What it stores
POTAFF	numeric	Potentially affected units
DO_NOT_DRIVE / PARK_O	UTSIDEstring	Urgent safety instructions for consumers

Table 1. Main fields used from the NHTSA recall dataset.

2.3 Checking the data quality

Before plotting, the dataset was checked for missing values, duplicate rows, inconsistent data types, and unrealistic derived values. Text fields were standardized to uppercase so that the same category would not be split only because of capitalization or whitespace. Date fields were converted to datetime format, while model year and potentially affected units were converted to numeric values. Exact duplicate rows were removed.

The most important cleaning decision was to create a campaign-level table. The raw file is not wrong, but it is more detailed than the campaign-level question we are asking. A single campaign can appear on many rows because it covers multiple models or years. Grouping by campaign number makes yearly counts, make rankings, and component rankings more meaningful.

Property	Value
Raw rows	240,137
Cleaned rows	240,137
Columns	23
Unique recall campaigns	14,912
Recall years	2010-2026
Unique makes	1,553

Unique components	590
Exact duplicate rows removed	0
Median affected units/campaign	474
90th percentile affected units	52,374
Do Not Drive campaigns	201
Park Outside campaigns	94

Table 2. Quality summary of the cleaned dataset.

3. Methodology

3.1 Software and reproducibility

All analysis was completed in Python. Pandas was used for data loading, cleaning, grouping, and summary tables. NumPy was used for log transformations and numeric calculations. Seaborn was used for the main visualizations, including line plots, bar plots, histograms, boxplots, heatmaps, and regression plots. SciPy was used for skewness, kurtosis, correlation, and the Shapiro-Wilk normality check.

3.2 Cleaning and campaign-level design

The workflow starts with the raw CSV, standardizes text fields, converts dates and numbers, removes exact duplicates, and then derives recall year, vehicle age at recall, log10 affected units, and special safety flag indicators. Text standardization was important because values with different capitalization or extra spaces could otherwise be treated as separate categories. Date conversion was also necessary so that recall-year trends and vehicle-age calculations could be created correctly.

The campaign-level table keeps the earliest recall date, representative make and component, maximum potentially affected population, model-year range, safety flags, and number of underlying records for each campaign. This step is important because the raw dataset is record-level, meaning one campaign can appear many times across different models or model years. If every row were counted as a separate recall, some makes and components would be overrepresented. Grouping by campaign number gives a more accurate view of unique recall activity.

3.3 Chart design choices

The visualizations were designed to be clear, readable, and appropriate for the type of data being shown. Bar charts were used for ranked category comparisons, such as top makes and top components. A line chart was used to show recall activity over time because it clearly displays year-to-year movement. Histograms and boxplots were used to study the distribution and spread of affected units, while the heatmap was used to compare component activity across years.

For affected-unit charts, a log10 scale was used because the raw values are highly skewed. Some campaigns affect only a small number of units, while others affect hundreds of thousands or more. Using the log scale prevents very large campaigns from visually dominating the chart and makes the overall pattern easier to interpret. Discrete palettes were used for category comparisons, while continuous palettes were used for heatmaps where color intensity represents higher or lower campaign counts.

3.4 Statistical checks

The affected-unit distribution was checked using descriptive statistics, skewness, excess kurtosis, and the Shapiro-Wilk normality test on a random sample of 5,000 recall campaigns. This step was important because the number of potentially affected units varies widely across campaigns. Some campaigns involve only a few vehicles, while others involve hundreds of thousands or even millions of units.

The raw affected-unit values are extremely right-skewed. The median campaign affects **474 units**, while the 90th percentile reaches **52,374 units** and the 99th percentile reaches **894,651 units**. This large difference between the median and upper percentiles shows that most campaigns are relatively small, but a small number of campaigns are very large. The raw affected-unit distribution has skewness of **23.86** and excess kurtosis of **965.88**, confirming a long right tail and the presence of extreme high-value campaigns.

Because of this skewness, a log10 transformation was applied to the affected-unit variable. After transformation, skewness drops from **23.86 to 0.30**, and excess kurtosis drops from **965.88 to -0.42**. This means the transformed distribution is much more balanced and easier to compare visually. The Shapiro-Wilk test also confirms that the raw values are not normally distributed, with a raw-sample p-value of **2.32e-92**. The log10-transformed values still do not perfectly follow a normal distribution, with a p-value of **6.68e-20**, but the distribution becomes much closer to a usable bell-shaped pattern for visualization and comparison.

Two relationship checks were also included. The Pearson correlation between approximate vehicle age and log10 affected units is **0.26**, suggesting only a weak positive relationship between vehicle age and campaign size. The Spearman correlation between model-year span and log10 affected units is **0.31**, which suggests that campaigns covering more model years tend to involve larger affected populations. Overall, these statistical checks support the use of log-scale visualizations and show that recall campaign size is driven by multiple factors, not by vehicle age alone.

4. Recall activity over time

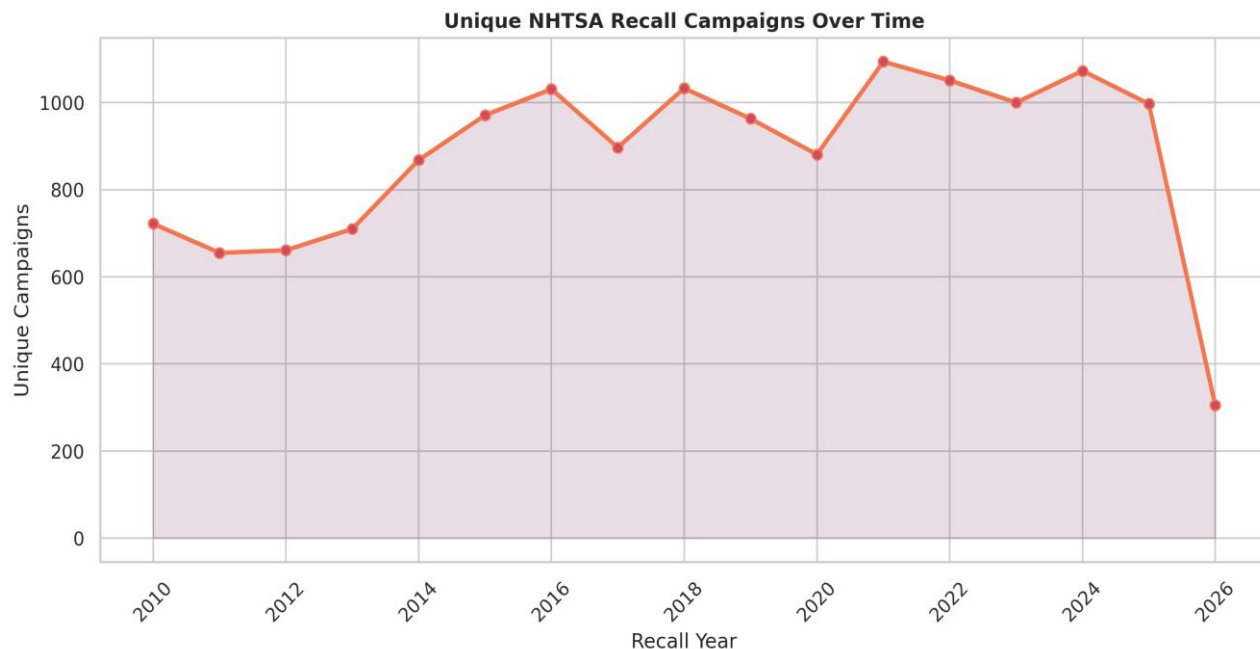


Figure 1. Unique NHTSA recall campaigns by recall year. The area under the line is shaded lightly to emphasize the overall time pattern.

Figure 1 shows that recall activity is not constant across the 2010-2026 period. The highest number of unique campaigns appears in 2021 with 1,094 campaigns, while the lowest appears in 2026 with 305 campaigns. This unevenness matters because a simple average would hide the fact that some years have much more recall activity than others.

The year-to-year movement may reflect several factors: changes in vehicle technology, regulatory attention, manufacturer reporting practices, and the timing of large recall investigations. Since this project is descriptive, the chart should be read as an activity pattern rather than proof of any one cause. A future project could join this data with production volume or vehicle sales to normalize recall activity by vehicles on the road.

5. Which makes dominate recall campaigns

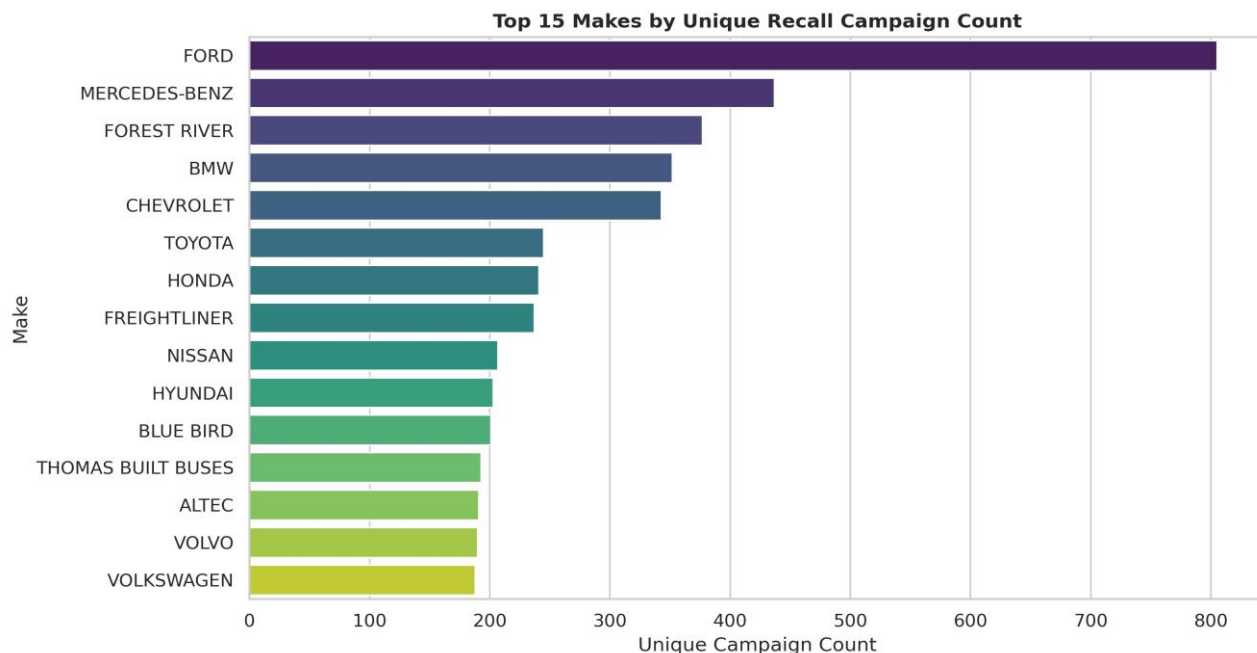


Figure 2. Top 15 makes by unique recall campaign count. Each bar is one make, and the chart uses campaign-level counts to reduce duplicate-row overcounting.

Figure 2 ranks the makes with the highest number of unique recall campaigns. The top make in this file is FORD, followed by MERCEDES-BENZ and FOREST RIVER. A high count does not automatically mean a make is less safe, because the dataset does not include how many vehicles were sold, how many models were active, or how large the manufacturer portfolio was. Larger brands can naturally have more recall campaigns because they sell more vehicles and have more model lines.

The useful takeaway is concentration: recall activity is not spread evenly across all makes. A small group of makes accounts for a large share of visible campaign activity. From an operations or customer-service perspective, this matters because manufacturers and dealers with higher campaign counts may need stronger recall communication systems, parts-planning processes, and repair scheduling capacity.

6. Which components dominate recall campaigns

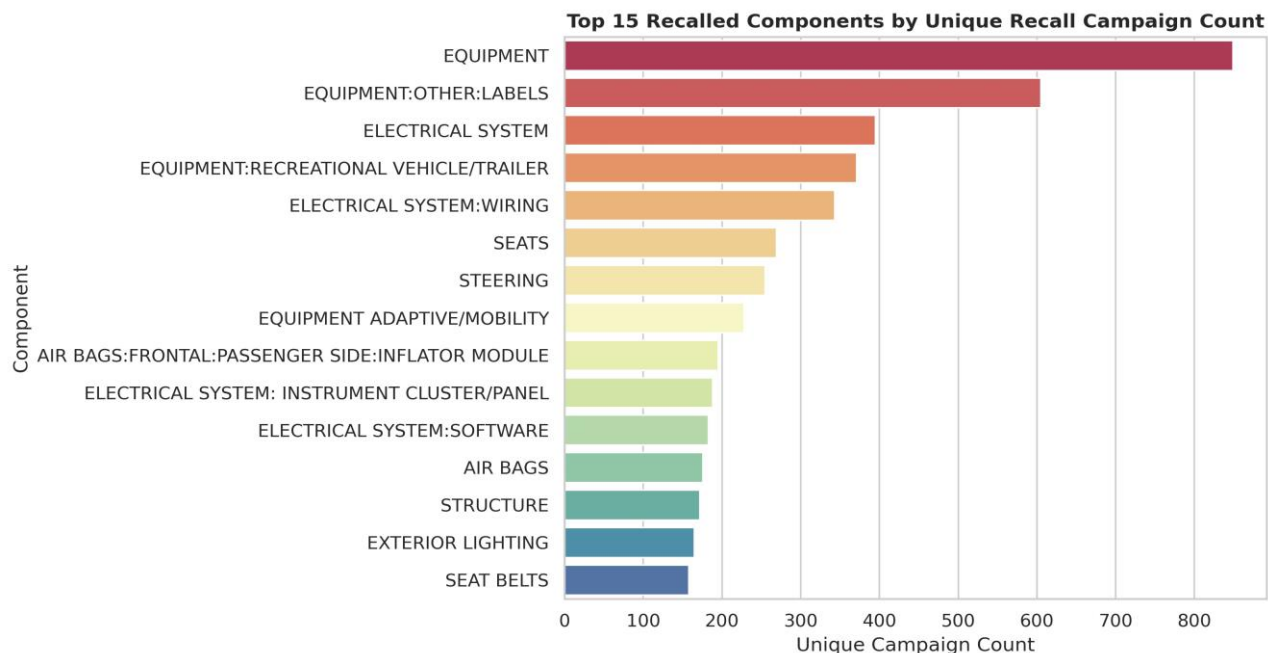


Figure 3. Top 15 component categories by unique recall campaign count.

Figure 3 shows that recall campaigns are also concentrated by component type. The most frequent component group is EQUIPMENT, followed by EQUIPMENT:OTHER:LABELS and ELECTRICAL SYSTEM. These categories appear often because they are broad systems that can include many specific subparts. For example, a component category may include mechanical parts, electrical controls, sensors, software, connectors, or supporting hardware.

This chart is helpful because it shows where recall attention is most commonly directed. Component groups with higher campaign counts can be viewed as priority areas for quality monitoring, supplier development, design reviews, and service readiness. However, the chart counts campaigns, not severity. A component with fewer campaigns could still be more urgent if those campaigns involve fire risk, crash risk, or immediate stop-use instructions.

7. How large recall campaigns are

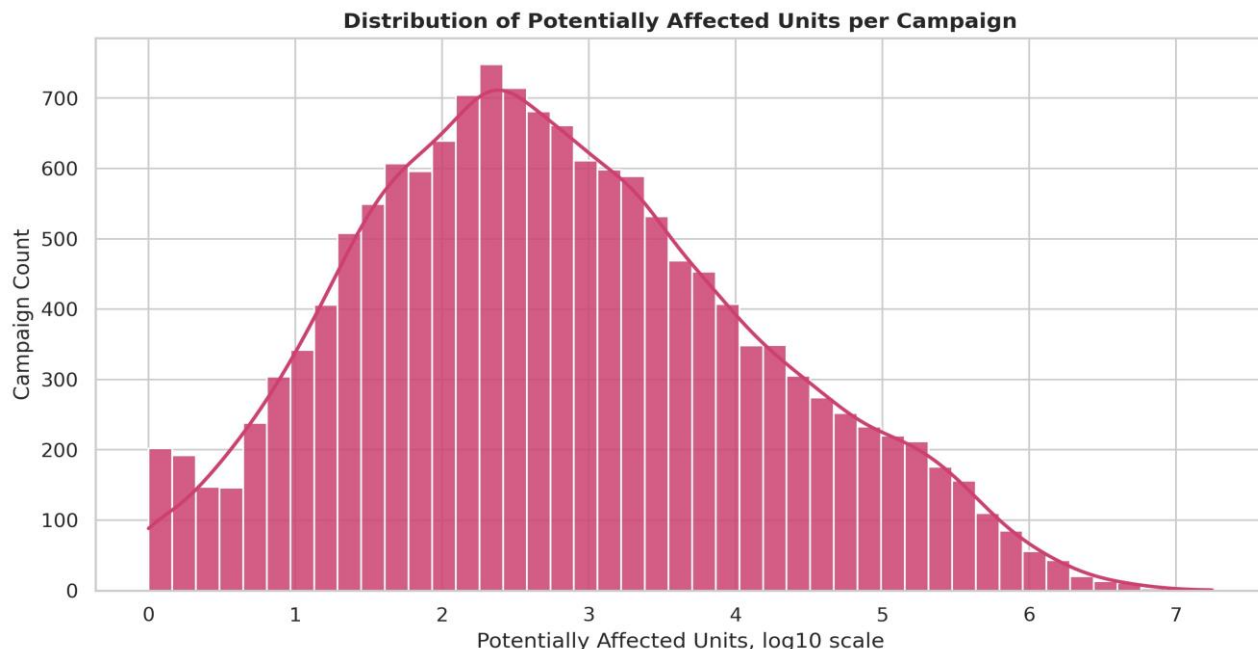


Figure 4. Distribution of potentially affected units per campaign on a log10 scale.

Figure 4 shows the distribution of potentially affected units per unique campaign. The raw distribution is strongly right-skewed: many campaigns affect a relatively small number of units, while a small number of campaigns affect very large populations. The median campaign affects 474 units, the 90th percentile reaches 52,374 units, and the 99th percentile reaches 894,651 units. Because the difference between small and large campaigns is so large, the log10 scale makes the pattern easier to see.

The shape of this distribution is important for planning. A typical campaign may require a manageable number of repairs, but the largest campaigns can create major pressure on parts availability, dealer labor capacity, customer communication, and warranty administration. This is why both the median and upper percentiles are reported. The median describes the typical campaign, while the high percentiles describe operational risk.

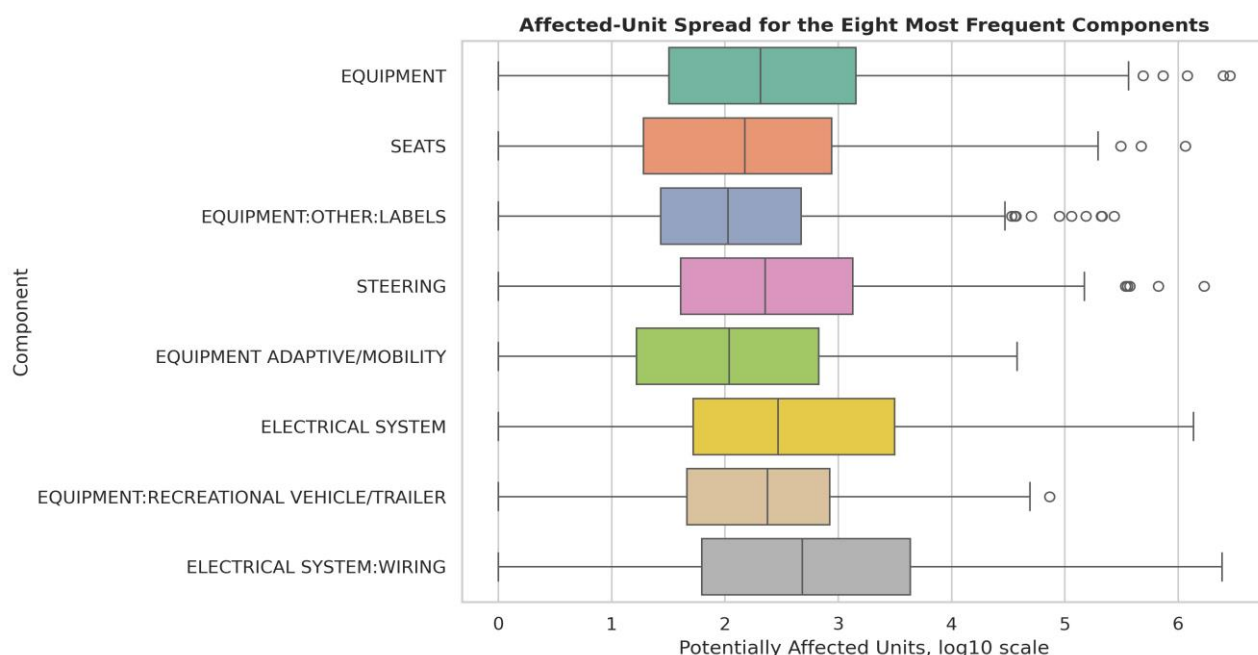


Figure 5. Affected-unit spread for the eight most frequent component groups.

Figure 5 compares campaign size across the most common component groups. The boxes show that even within the same component group, campaign size varies widely. Some categories have long upper tails, meaning a few campaigns within those groups affect many more units than the typical campaign. This reinforces the point that campaign count and campaign scale are different measures: a component can appear frequently but mostly in small campaigns, or appear less often but include a few very large recalls.

7.1 Component concentration by year

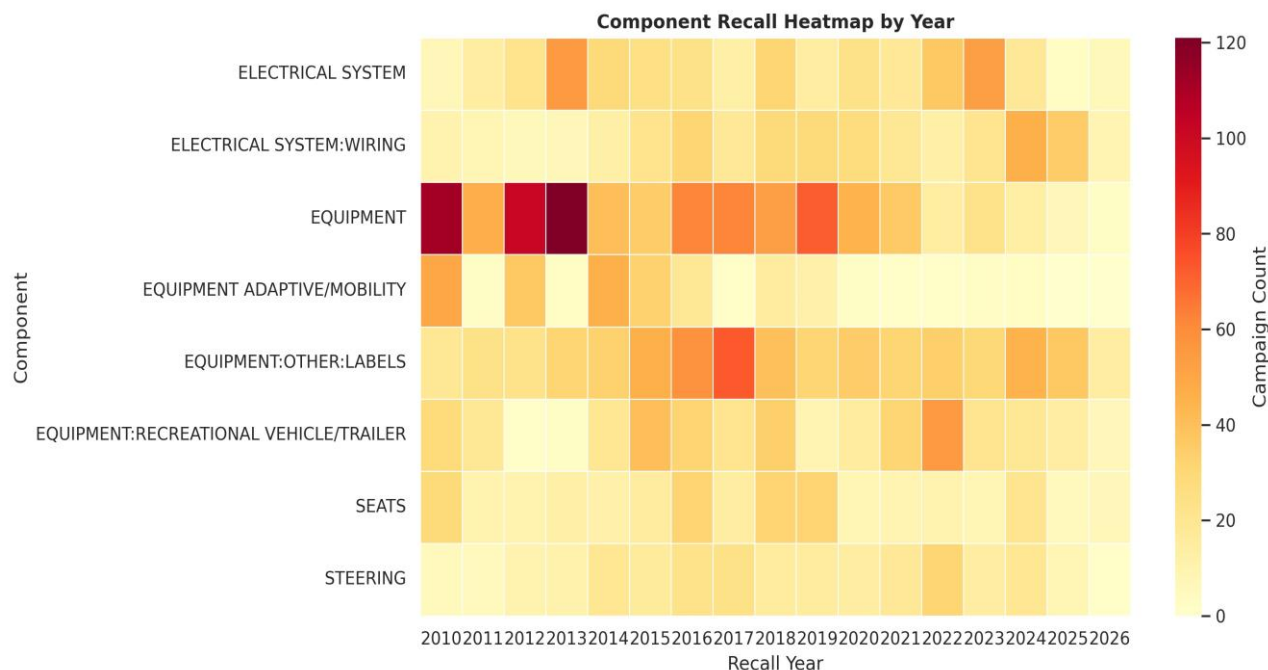


Figure 6. Heatmap of frequent component groups by recall year. Darker red cells indicate higher campaign counts.

Figure 6 adds a time dimension to the component analysis. Instead of only asking which components are most common overall, the heatmap shows when those component groups were more active. Darker cells indicate higher campaign counts in a specific component-year combination. This view is useful because component concerns may rise or fall over time as vehicle technology changes, suppliers change, or regulators focus on different safety issues.

The heatmap is also a good example of why Seaborn is useful for advanced static visualization. A table of component-year counts would be hard to read quickly, but the color gradient immediately highlights clusters. For a follow-up analysis, the darkest cells could be investigated one by one to identify the specific recall campaigns driving those peaks.

8. Urgency flags and influence sources

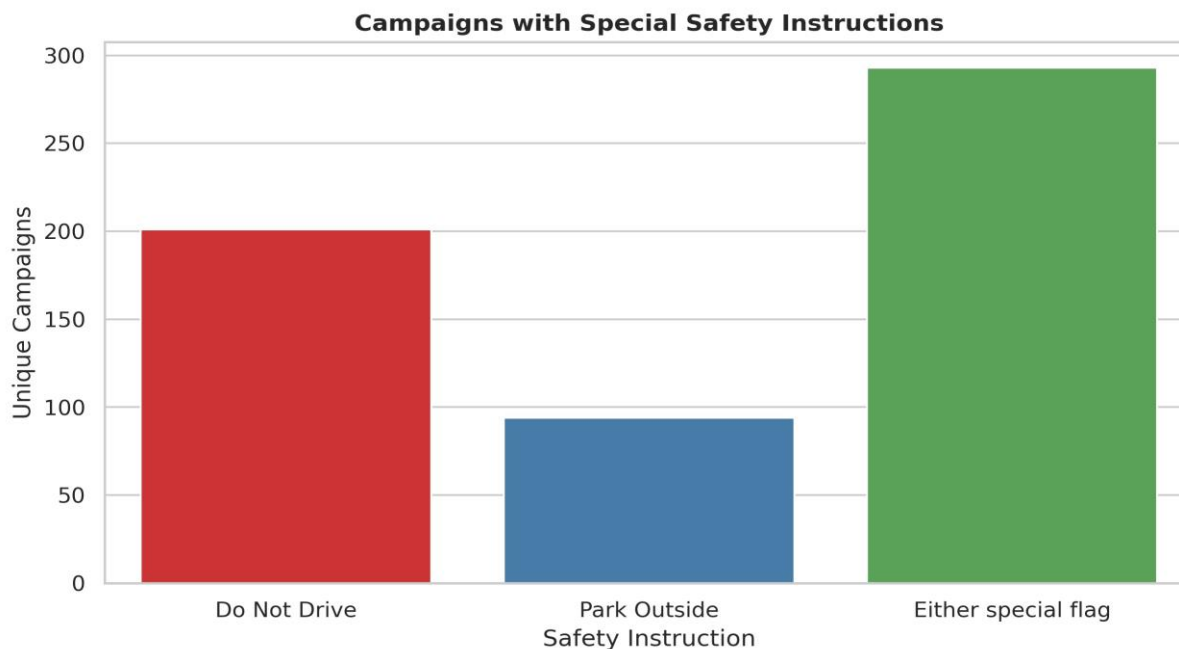


Figure 7. Campaigns with special safety instructions.

Figure 7 summarizes high-urgency consumer instructions. The campaign-level table contains 201 Do Not Drive campaigns and 94 Park Outside campaigns. The combined special-flag count is 293. These records are especially important because they tell the owner to change behavior immediately, not just schedule a repair when convenient.

The Park Outside flag is commonly associated with risks that may continue even when the vehicle is not being driven, such as fire-related concerns. The Do Not Drive flag signals that operating the vehicle before repair may create unacceptable risk. From a communication perspective, these campaigns require clearer messaging and faster customer response than routine recall notices.

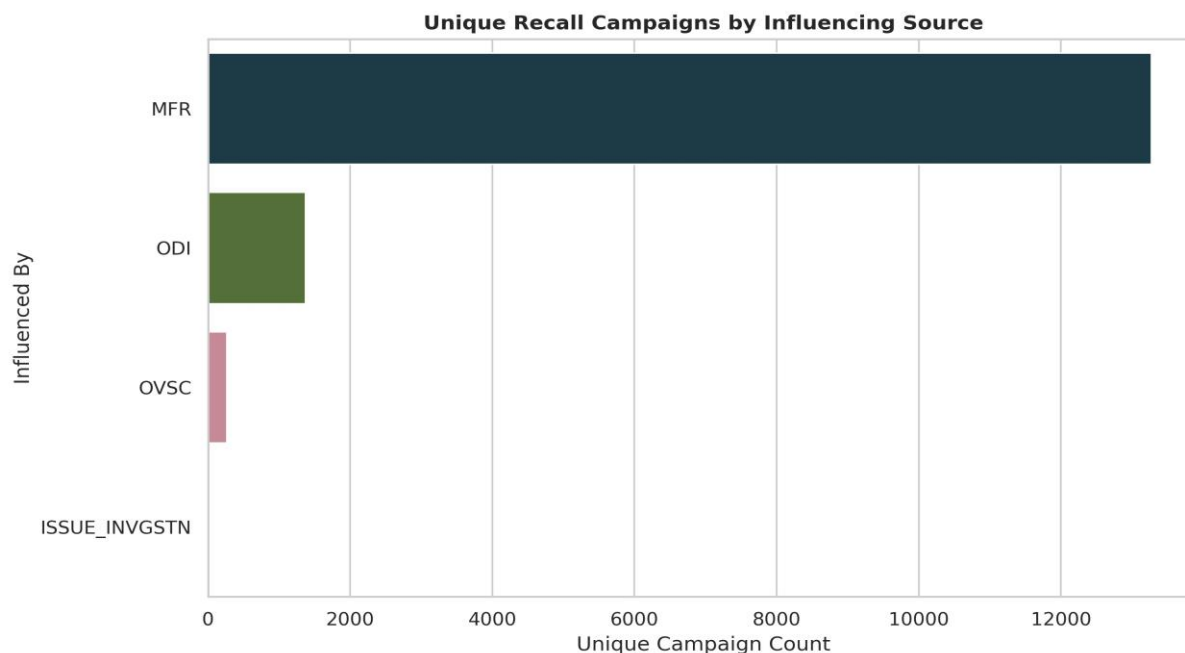


Figure 8. Unique recall campaigns by influencing source.

Figure 8 shows how recall campaigns are distributed by the field labeled `influenced_by`. The largest category in this file is MFR, which means most recall campaigns were influenced or initiated by the manufacturer. This suggests that manufacturers play the main role in identifying, reporting, and managing recall campaigns in the dataset. The second major category is ODI, which refers to the Office of Defects Investigation. Although ODI has a much smaller count than MFR, it is still important because it may reflect recalls connected to regulatory review, defect investigations, consumer complaints, or safety monitoring activities.

The smaller categories, such as OVSC and ISSUE_INVGSTN, appear far less often, showing that recall activity is not evenly distributed across all influence sources. However, this chart should be interpreted carefully because the `influenced_by` field does not fully explain the cause, severity, or responsibility of a recall. A manufacturer-influenced recall is not automatically less serious, and an investigation-influenced recall is not automatically more serious. The chart is best used as a general view of recall source patterns rather than a final judgment about recall responsibility.

9. Vehicle age and campaign scale

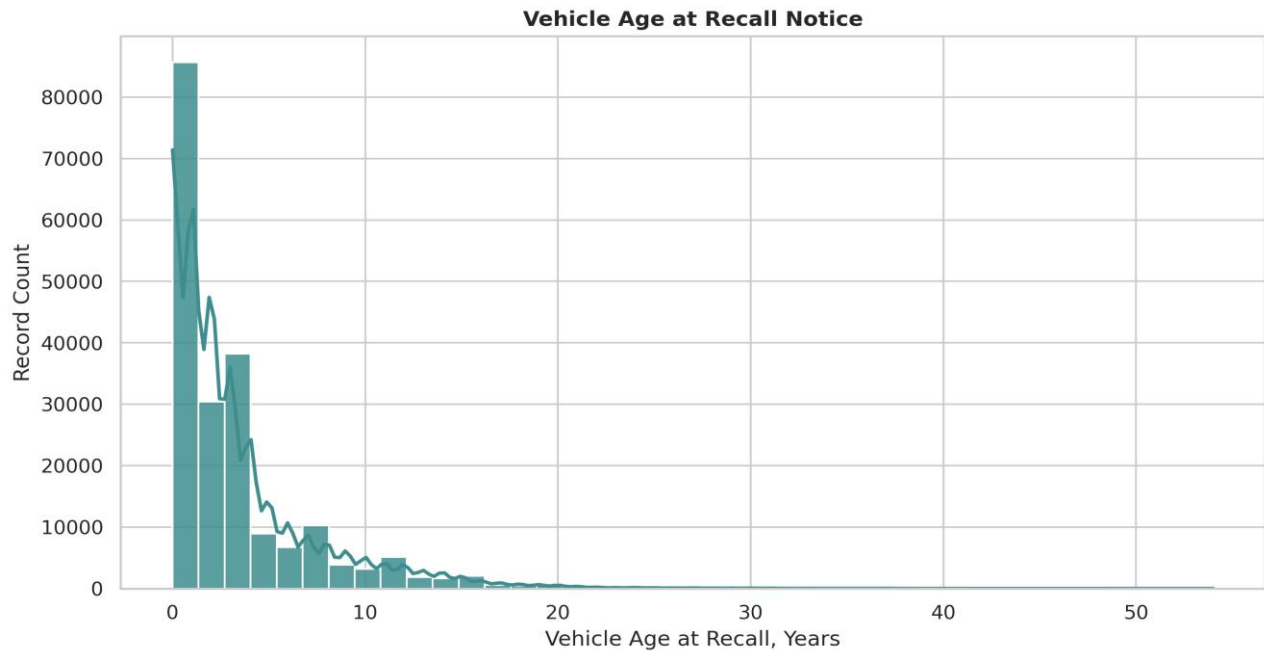


Figure 9. Distribution of vehicle age at recall notice.

Figure 9 uses record-level data to estimate how old vehicles are when recall notices appear. Many recall records occur close to the model year, but the distribution also includes older vehicles. This is expected because some defects are identified after vehicles have been in service for several years. Older-vehicle recalls can be harder to complete because vehicles may have changed owners, moved to different regions, or left dealer-service networks.

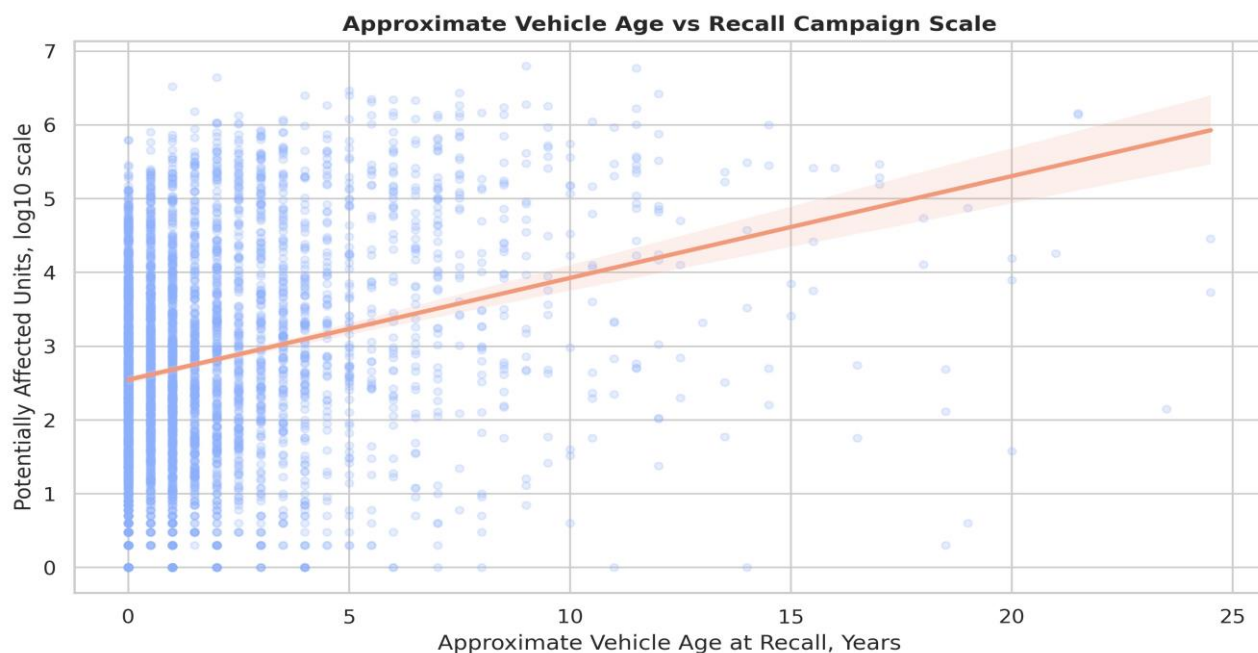


Figure 10. Approximate vehicle age at recall versus affected campaign scale. Points are sampled for readability, and the trend line is fitted on the log-scale affected-unit value.

Figure 10 checks whether approximate vehicle age is strongly related to campaign scale. The Pearson correlation between vehicle age and log10 affected units is 0.26, which suggests a weak relationship in this file. In plain language, older recalls are not automatically larger or smaller. Campaign scale appears to depend more on the number of models and production volume involved than on age alone.

The Spearman correlation between model-year span and log10 affected units is 0.31. This is more meaningful because campaigns covering more model years often include more affected units. Still, this is only a simple relationship; a stronger model would need manufacturer, make, component, sales volume, and campaign type as additional predictors.

10. What this analysis cannot tell us

There are several limitations. First, the dataset counts recall records and campaigns, but it does not include vehicle sales or exposure. A make with many campaigns may simply have a larger market presence. Second, component names are broad, so a single component category can contain many different technical problems. Third, potentially affected units is a maximum campaign-level value in this analysis, but the final number of repaired vehicles may be lower because recall completion depends on customer response and service execution.

Another limitation is that the analysis is descriptive. It shows patterns, but it does not prove causes. For example, a year with a high number of campaigns could reflect more defects, more reporting, a few large investigations, or changes in recall practices. Similarly, a component with many campaigns is not automatically the most dangerous component. Severity, consumer risk, and repair urgency require additional fields and possibly reading the detailed recall descriptions.

10.1 Conclusions and next steps

This report shows that the NHTSA recall dataset can be turned into a clear Seaborn-based visual analysis by cleaning the raw records and summarizing them at the campaign level. The key findings are: recall activity varies by year, campaign counts are concentrated among a smaller set of makes and components, affected-unit counts are highly right-skewed, and special safety flags identify a smaller but important group of urgent campaigns.

The most important technical lesson is the need to analyze the data at the correct level. Record-level rows are useful for model-year and vehicle-age analysis, but campaign-level grouping is better for recall counts and affected-unit summaries. Without this step, repeated rows could make some categories look more active than they really are.

Three next steps would improve the project. First, add vehicle sales or registration data so recall rates can be normalized by market exposure. Second, read and classify the detailed recall descriptions to separate risk types such as fire, crash, airbag, software, or labeling. Third, build a predictive model to estimate which campaign characteristics are associated with very large affected-unit populations or urgent safety flags.

11. References

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12. Appendix A. Detailed numerical results

A.1 Top makes

Make	Campaign Count
FORD	805
MERCEDES-BENZ	437
FOREST RIVER	377
BMW	352
CHEVROLET	343
TOYOTA	245
HONDA	241
FREIGHTLINER	237
NISSAN	207
HYUNDAI	203

A.2 Top components

Component	Campaign Count
EQUIPMENT	849
EQUIPMENT:OTHER:LABELS	605
ELECTRICAL SYSTEM	395
EQUIPMENT:RECREATIONAL VEHICLE/TRAILER	371
ELECTRICAL SYSTEM:WIRING	343
SEATS	269
STEERING	255
EQUIPMENT ADAPTIVE/MOBILITY	228
AIR BAGS:FRONTAL:PASSENGER SIDE:INFLATOR MODULE	195
ELECTRICAL SYSTEM: INSTRUMENT CLUSTER/PANEL	188

A.3 Statistical summary

Measure	Value
Raw affected-unit skewness	23.86
Log10 affected-unit skewness	0.30
Raw affected-unit excess kurtosis	965.88
Log10 affected-unit excess kurtosis	-0.42
Shapiro p-value, raw sample	2.32e-92
Shapiro p-value, log10 sample	6.68e-20
Pearson r: vehicle age vs log campaign scale	0.26
Spearman rho: model-year span vs log scale	0.31